Overview

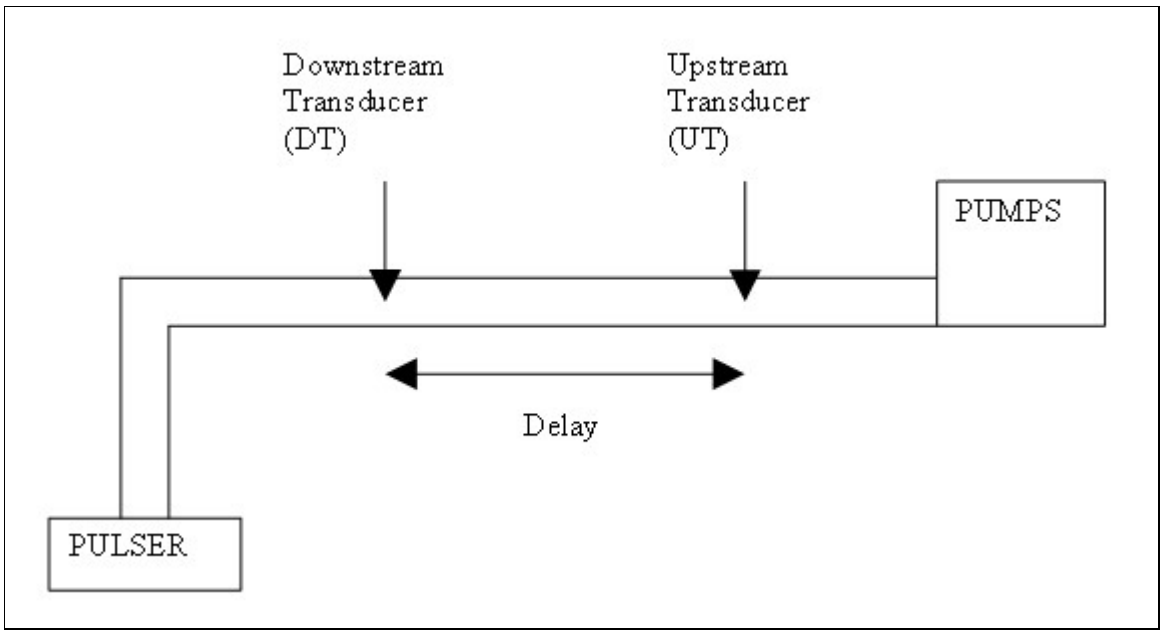
You are in the SARA II Configurator application section.

# Dual Pressure Transducer Algorithm

## DPT Principle of Operation

The Dual Pressure Transducer (DPT) algorithm essentially acts as a directional signal amplifier. When configured correctly, upstream traveling components of the signal will be amplified while downstream traveling components of the signal will be attenuated.

Its principle of operation is very simple:



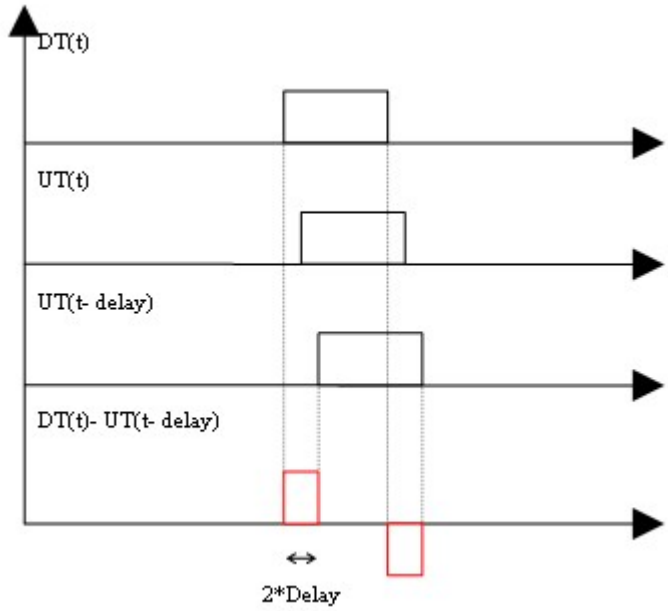

The key parameter is the delay between signals from Upstream Transducer and Down Stream Transducer.

The DPT algorithm estimates the delay within a range of physically possible values and then performs the following operation:

```
DPT output(t)= DT(t)- UT(t- delay)
```

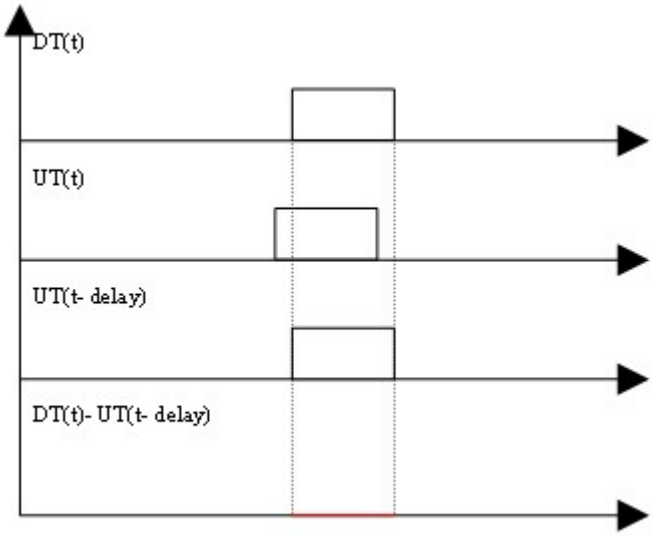
Figures 1 and 2 below illustrate the difference between MWD pulser and mud pump:

(i) Pulser Impulse.




If you look at the difference signal, you can observe that it is close to the time derivative of the Input Impulse. By integrating it you will get (as the Clean Output of the DPT algorithm) a pulse close to the original shape.

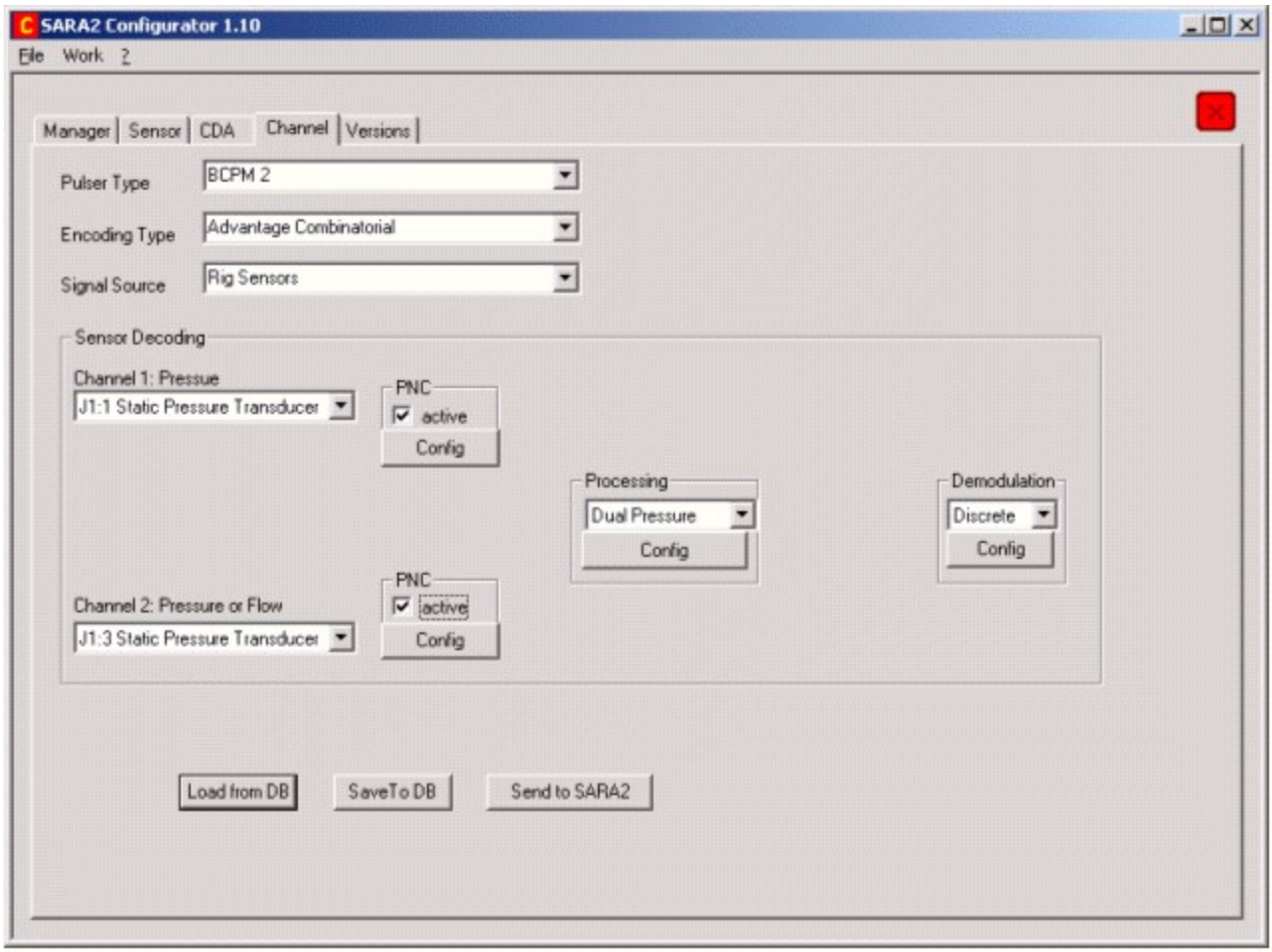
(ii) Pump Impulse.




If you look at the difference signal you can see that it is zero. That means that the Pump Signal (in an ideal case) does not pass through the DPT, while the pulser signal passes normally.

## DPT Setup with the SARA II Software

The SARA 2 Configuration software offers the following parameters for the DPT algorithm.




Click the **Channel** tab to bring it forward. The **Sensor Decoding** section of the window offers the following options:

- **Channel 1: Pressure**: Select the DPT sensor located nearest to the pumps.
- **Channel 2: Pressure or flow**: Select the DPT sensor located nearest to the pulser.

Select **Dual Pressure** from the **Processing** drop-down menu and click the **Config** button.

- **Default Delay (ms)**: Enter the estimated delay between both transducers. Calculate this as follows:

```
Delay= pipe_length/wave_speed
```

- **Minimum Delay (ms)**: minimum delay of the wave between both transducers.
- **Maximum Delay (ms)**: maximum delay of the wave between both transducers.
- **Correlation Length(s)**: In the SARA 2, this is currently fixed at four seconds. We will investigate if changing the correlation length and the correlation update frequency improves the results of the algorithm. This correlation is currently updated every two seconds.

These parameters will be reduced to one (the distance between transducers) for the field version of the algorithm.

Typical values for the parameters are:

* Default Delay [milliseconds] = 1.0 * distance[meters]

* Maximum Delay [milliseconds] = 1.3 * distance[meters]

* Minimum Delay [milliseconds] = 0.6 * distance[meters]

---

**Note:** As mentioned above, **Correlation Length** is fixed to 4 seconds. Therefore, whatever you set will be ignored.

---

See also:

- DPT Recommendations
- Rig Setup